



Unit 4 · Outcome 3 · Scientific Investigation

Findings & implications

KK12

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



The "so what?" moment

SECTION 01 The "so what?" moment An investigation that stops at a graph is only half-finished. The marks live in two sentences: what you found , and why it matters . Imagine you spend a weekend baking, taste-testing and recording which oven shelf browns a cake most evenly. If a friend asks "how did it go?" and you just hand them a spreadsheet of temperatures, they'll stare at you. What they want is the finding ("the middle shelf browns most evenly") and the implication ("so put cakes on the middle shelf — and it probably applies to biscuits too"). Scienc

Result → finding → conclusion → implication

SECTION 02 Result → finding → conclusion → implication These four words are not synonyms, and examiners deliberately test whether you can tell them apart. Think of them as a relay — each one passes the baton a step further from the raw numbers. Term What it is Priya's creek, worked Result A raw, specific number or observation straight from your data — no interpretation. "Mean turbidity at 0 m was 41 NTU; at 175 m it was 12 NTU." Finding The pattern or relationship the results show , in plain interpreted language. "Turbidity decreased steadily with distan

Term	What it is	Priya's creek, worked
Result	A raw, specific number or observation straight from your data — no interpretation.	"Mean turbidity at 0 m was 41 NTU; at 175 m it was 12 NTU."
Finding	The pattern or relationship the results show , in plain interpreted language.	"Turbidity decreased steadily with distance downstream of the outfall."
Conclusion	The answer to the aim/hypothesis, stated with the level of support the evidence gives.	"The data supports the hypothesis: the stormwater outfall raises turbidity in Wirrim Creek nearest the outfall."
Implication	The "so what?" — meaning beyond this study: for management, other sites, theory, or future work.	"A sediment trap at the outfall could reduce downstream turbidity; similar outfalls on urban creeks may need monitoring."

Writing a key finding

SECTION 03 Writing a key finding A key finding is the headline relationship your investigation uncovered. A strong one does three things at once: Names the relationship between your independent and dependent variables (what changed with what). Uses the direction word — increased, decreased, was higher/lower, showed no difference. Anchors it in quantified evidence — a value, a range, or a comparison from your own data. Finding formula As [independent variable] [changed], [dependent variable] [direction] (from X to Y), indicating [relationship]. e.g. "As d

The implication radius

SECTION 04 The implication radius Implications spread outward from a finding like ripples. The trick is to recognise the different directions meaning can travel — examiners reward breadth here. Explore the five domains below. click a domain · explore the implications SELECT A DOMAIN Click a glowing node to see how Priya's single finding ripples into that area of meaning. Direction Question it answers Priya's creek Environmental What does this mean for the ecosystem itself? High turbidity reduces light, harming aquatic plants and sediment-sensitive macroi

Direction	Question it answers	Priya's creek
Environmental	What does this mean for the ecosystem itself?	High turbidity reduces light, harming aquatic plants and sediment-sensitive macroinvertebrates near the outfall.
Management	What action could follow?	Install a sediment trap or revegetate the bank to filter stormwater before it reaches the creek.
Generalisation	Where else might this apply?	Other urban creeks with stormwater outfalls may show the same gradient — worth monitoring.
Scientific / theory	How does it fit known science?	Supports the model that urban runoff is a point source of suspended sediment, consistent with catchment hydrology.
Further research	What should be investigated next?	Sample across seasons or during rain events to test whether the effect intensifies after storms.

Staying inside your evidence

SECTION 05 Staying inside your evidence The most common way to lose implication marks is to over-reach — to claim more than your data can support. A finding from one creek, on one day cannot prove a fact about every waterway on Earth . The skill is matching the scope of your claim to the scope of your evidence . Scope-o-meter · how far can this claim travel? Drag the slider to widen Priya's claim. Watch where the evidence stops supporting it. Three over-reach phrases examiners pounce on "This proves ..." — a single student investigation supports or is cons

Command words

◆ COMMAND WORDS

SECTION 06 Command words The verb in the question controls how much "so what?" you owe. Tap each to reveal what it demands.

Exam traps

⚠ EXAM TRAPS

SECTION 07 Exam traps The recurring ways students leak marks on findings-and-implications questions:

Writing trainer · P·E·L

P·E·L WRITING

SECTION 08 Writing trainer · P·E·L Strong implication answers follow P·E·L — P oint (state the implication), E vidence (cite your own data), L ink (connect to management, theory or the wider world). Draft a response below; the meters light up as you hit each move. Prompt · "State one key finding from Priya's study and discuss one implication of it. (4 marks)" P · Point 0% State the finding clearly. E · Evidence 0% Quote a value or comparison. L · Link 0% Reach outward — so what?



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Weak vs strong

✗ WEAK ANSWER

✗ WEAK — 1 / 4 "Priya found that the turbidity was high. This proves that all stormwater drains pollute every river , so the council should ban all drains ." No number or relationship — "high" is not a finding, it's vague. "Proves... all rivers" massively over-reaches one creek on one day. The implication ("ban all drains") is not justified by the evidence and isn't realistic management.

✓ STRONG ANSWER & MARKING

✓ STRONG



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice exam

★ PRACTICE & SELF-MARK

SECTION 10 Practice exam Five graded questions, 20 marks total. Attempt each, reveal the marking scheme, then self-mark — your score collects in the dock.

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Priya found that the turbidity was high. This proves that all stormwater drains pollute every river , so the council should ban all drains ." No number or relationship — "high" is not a finding, it's vague. "Proves... all rivers" massively over-reaches one creek on one day. The implication ("ban all drains") is not justified by the evidence and isn't realistic management.

✓ STRONG / MODEL ANSWER

A key finding was that mean turbidity decreased from 41 NTU at the outfall to 12 NTU at 175 m downstream , indicating the outfall is the main source of suspended sediment. Finding = relationship + direction + quantified evidence. ✓ One implication is that installing a sediment trap or revegetating the bank could reduce turbidity and protect light-sensitive aquatic life near the outfall. Implication links the finding to a realistic management action and an ecological consequence. ✓ As only one creek on one day was sampled, this suggests rather than proves the effect, and similar urban creeks would warrant monitoring ." Hedged scope + a generalisation/ further-research implication. ✓